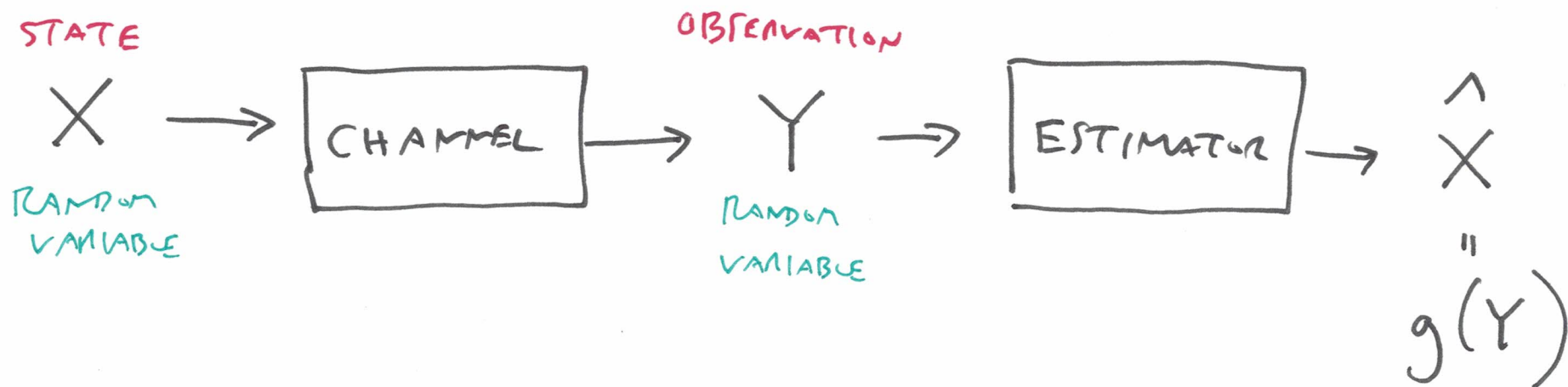


ESTIMATION PROBLEMS

SETUP:



GOAL: MAKE $g(Y)$ AS CLOSE AS POSSIBLE TO THE ORIGINAL INPUT X .

OR FIND THE **BEST** ESTIMATE OF X GIVEN Y .

3 TYPES OF ESTIMATION WE CARE ABOUT:

- ML ESTIMATION (MAXIMUM LIKELIHOOD)
 - ↳ PRIORS ARE NOT INVOLVED (AGNOSTIC/
UNINFORMATIVE PRIOR)
- MAP ESTIMATION (MAXIMUM A POSTERIORI)
 - ↳ TAKE PRIORS INTO ACCOUNT
- MMSE ESTIMATION (MINIMUM MEAN
SQUARE ESTIMATION)
 - ↳ LINEAR MMSE

CONSIDER JOINT PMF
Y

		1	2	3	4	
X	-1	0.04	0.04	0.08	0.24	→ 0.4
	0	0.1	0.1	0.15	0.15	→ 0.5
	1	0.04 ↓ 0.18	0.04 ↓ 0.18	0.01 ↓ 0.24	0.01 ↓ 0.4	→ 0.1

ESTIMATOR:

$$\hat{X} = g(Y).$$

ML ESTIMATION: GIVEN THE VALUE
OF Y , WHAT VALUE OF X IS
MOST LIKELY TO HAVE PRODUCED IT?

$$\hat{X} = g_{ML}(Y) = \arg \max_x f_{Y|X}(y|x)$$

THE VALUE
OF X THAT
MAXIMIZES...

CREATE

$f_{Y|X} :$

$X \backslash Y$	1	2	3	4
-1	0.1	0.1	0.2	0.6 → 1
0	0.2	0.2	0.3	0.3 → 1
1	0.4	0.4	0.1	0.1 → 1

BIGGEST

IN

COLUMN

$$\hat{g}_{ML}(Y) = \begin{cases} 1 & Y = 1 \text{ or } 2 \\ 0 & Y = \cancel{2} 3 \\ -1 & Y = 4 \end{cases}$$

NORMALIZED
ROWS

NOTE THAT ML ESTIMATE IS CHOOSING
 $X = 1$ WHICH IS NOT THAT LIKELY.

BETTER: MAP ESTIMATE THAT
TAKES PRIORS INTO ACCOUNT.

GIVEN THE VALUE OF Y , WHAT
VALUE OF X IS MOST LIKELY
TO HAVE OCCURED?

$$\hat{X} = g_{\text{MAP}}(Y) = \arg \max_x f_{X|Y}(x|y)$$

CREATE

$f_{X|Y}$:

(*) CAN JUST READ OFF
THE JOINT PMF TABLE

$X \backslash Y$	1	2	3	4
-1	0.22	0.22	0.33	0.6
0	0.56	0.56	0.63	0.38
1	0.22	0.22	0.04	0.02

BIGGEST IN
COLUMN

↓
ADD TO 1

$$g_{MAP}(Y) = \begin{cases} 0 & Y=1, 2, 3 \\ -1 & Y=4 \end{cases}$$

PROBABILITIES OF ERROR:

ADD UP PROBABILITIES IN JOINT PMF
TABLE THAT ARE NOT CHOSEN.

ML ERROR:

$$0.04 + 0.04 + 0.1 + 0.1 + 0.08 + 0.01 + 0.15 + 0.01$$

$$= 1 - (0.04 + 0.04 + 0.15 + 0.24)$$

$$= 1 - (0.47) = \boxed{0.53}$$

$$\begin{array}{r} 0.24 \\ 0.15 \\ 0.04 \\ 0.01 \\ \hline 0.47 \end{array}$$

MAP ERROR:

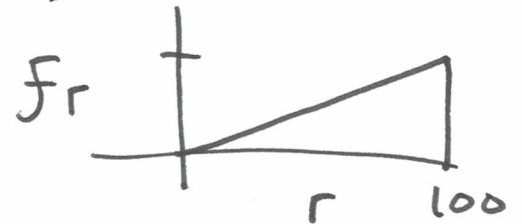
$$1 - (0.1 + 0.1 + 0.1 + 0.24) = 1 - 0.54 = \boxed{0.46}$$



RECEIVED POWER MEASUREMENT IN dB

$$X = Y - (40 + 40 \log_{10} R)$$

Y IS GAUSSIAN $\mu=0, \sigma=8$



R HAS PDF: $f_R(r) = \begin{cases} \frac{1}{5000} & r \in [0, 100] \\ 0 & \text{ELSE} \end{cases}$

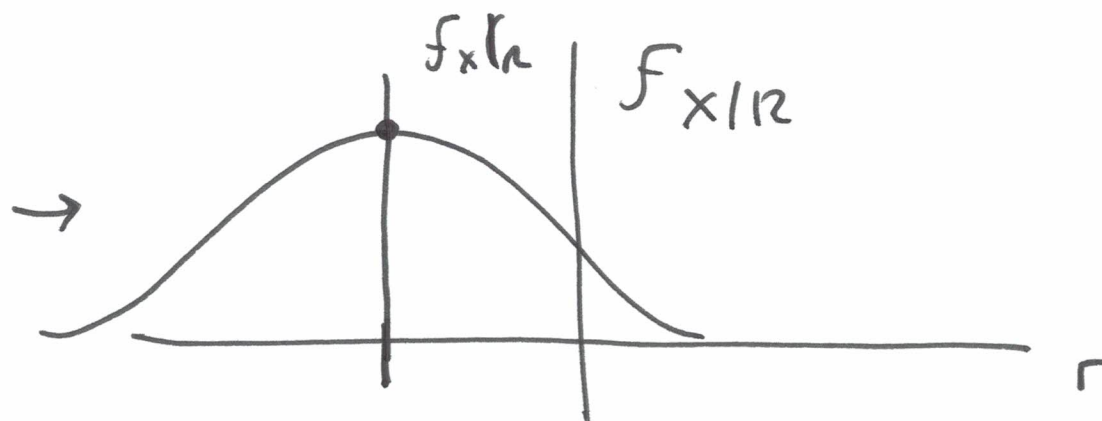
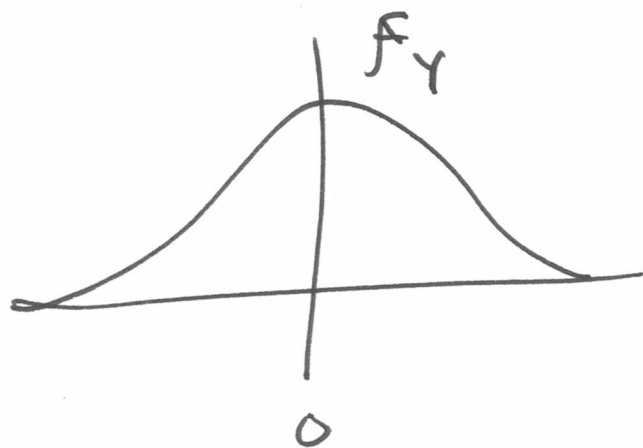
GOAL: ESTIMATE R
GIVEN X .

WHAT IS ML ESTIMATE R GIVEN X ?

WE NEED $f_{X|R}$. WHAT IS THIS PDF?

IF R IS FIXED, THEN X IS THE

GAUSSIAN PDF OF Y SHIFTED OVER:



$$\mu_{X|R} = -40 - 40 \log_{10} R.$$

THIS PDF IS MAXIMIZED AT THE MEAN.

OBSERVATION

$$X = -40 - 40 \log_{10} \hat{\Gamma} \quad \text{TO ESTIMATE}$$

$$10^{\frac{X+40}{-40}} = \hat{\Gamma} = 10^{-\frac{X}{40} - 1} = \boxed{\frac{1}{10} 10^{-\frac{X}{40}}} \quad \hat{\Gamma}_{ML}$$

$$f_{X|R} = \boxed{m} e^{-\frac{(X+40+40 \log_{10} \Gamma)^2}{2 \cdot 8^2}}$$

WHAT ABOUT THE MAP ESTIMATE?

WE DIDN'T USE PRIOR INFO ABOUT τ .

WE NEED $f_{R|X} = \frac{f_{X|R} f_R}{f_X} = \frac{f_{RX}}{f_X}$

$f_{X|R} f_R = \boxed{\text{const.}} \downarrow \frac{f_R}{5000} e^{-\frac{(x+40+40 \log_{10} r)^2}{128}}$

WHAT r MAXIMIZES THIS?

$$\frac{d}{dr} = 0.$$

$$e^{-\boxed{}} - r^{\frac{1}{64} (x + 40 + 40 \log_{10} r)} e^{-\boxed{}} = 0.$$

o
o
o

$$1 - \frac{5}{8} \log_{10} e (x + 40 + 40 \log_{10} r) = 0$$

$$\Rightarrow \hat{r}_{\text{MAP}} = (0.1236) 10^{-\frac{1}{40} x}$$

SAME FORM BUT ESTIMATES r TO BE
BIGGER DUE TO THE PRIOR.

LAST IDEA: MINIMUM MEAN
SQUARE ESTIMATION (MMSE):

$$g_{\text{MMSE}}(Y) = E(X|Y)$$

RESULTS FROM MINIMIZING

$$e = E \left((g(Y) - X)^2 \right)$$

THIS MAY NOT BE AN INTEGER.

IN OUR EXAMPLE,

$$g_{MMSE}(Y=1) =$$

$$-1 \cdot \frac{0.04}{0.18} + 0 \cdot \frac{0.1}{0.18} + 1 \cdot \frac{0.04}{0.18}$$

↑

↑

↑

$$\sum_{X=-1,0,1} X \cdot f_{X|Y}$$

$$g_{MMSE}(Y) =$$

$\left\{ \begin{array}{l} \text{NOT} \\ \text{INTEGER} \end{array} \right.$

$Y=1, 2$

$Y=3$

$Y=4$

SOMETIMES WE WANT A CLEANER
ANSWER SINCE

$E(X|Y)$ CAN BE A MESSY
INTEGRAL

IN THIS CASE, WE CAN USE THE
LINEAR MMSE

FIND a AND b SO THAT WE MIN

$$E \left(\underbrace{(aY + b) - X}_{} \right)^2$$

WON'T BE AS GOOD AS FULL MMSE.
SIMPLE LINEAR FUNCTION OF Y .

IT TURNS OUT THAT

$$a = \rho_{xy} \frac{\sigma_x}{\sigma_y}$$

$$b = \mu_x - a\mu_y$$



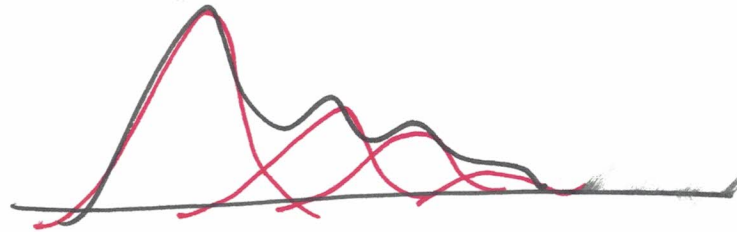
END OF CLASS CONTENT!!

WHERE TO GO FROM HERE?

- APPLICATIONS IN ML.

- PARAMETER ESTIMATION

- DENSITY ESTIMATION (PDF ESTIMATION)

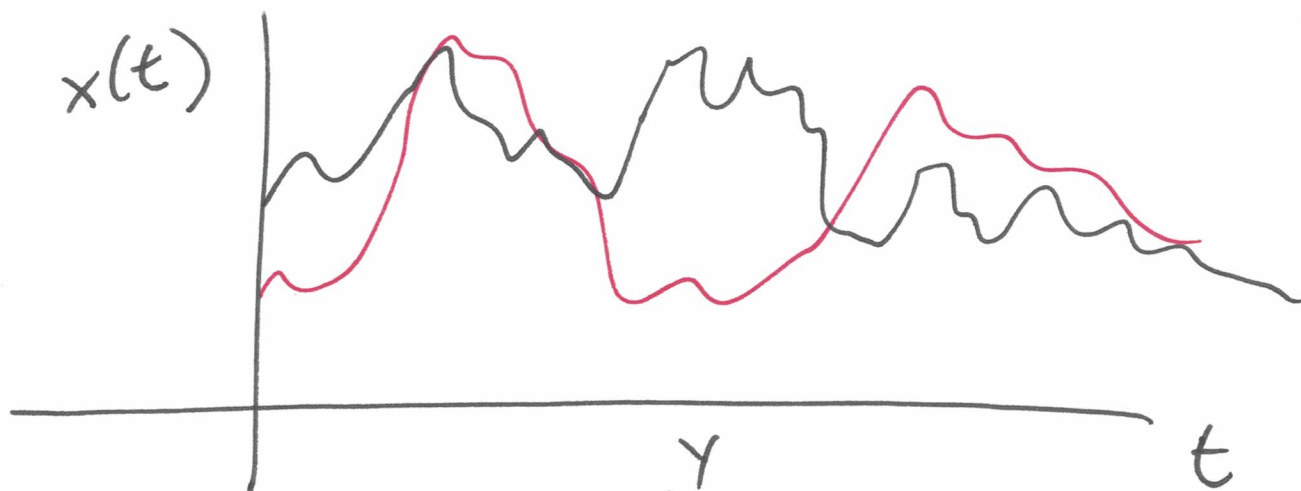


RANDOM VECTOR

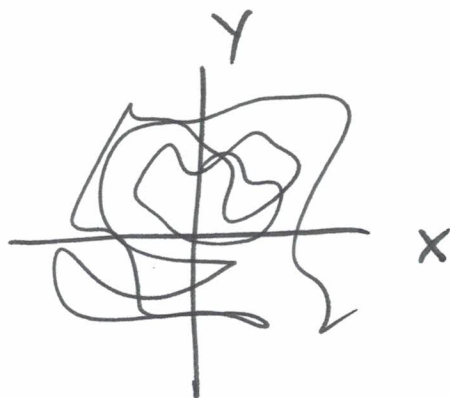
$$\begin{bmatrix} x_1 \\ \vdots \\ x_N \end{bmatrix}$$

, μ, Σ

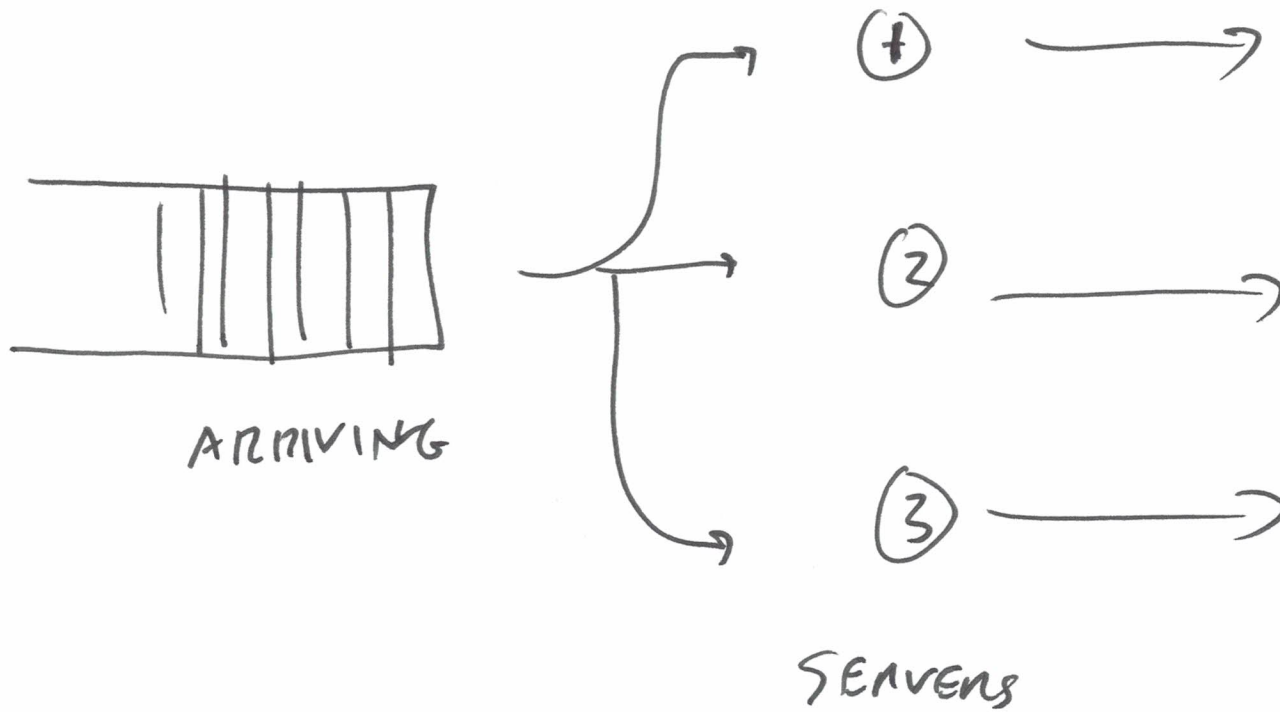
RANDOM PROCESS



2D RANDOM
PROCESS



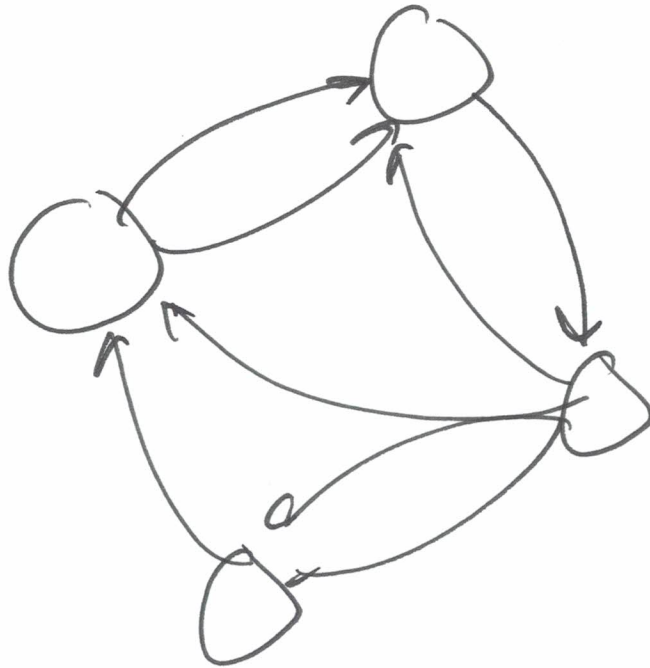
QUEUEING THEORY:



MARKOV PROCESS:

X_{t+1} DEPENDS ON

$X_t, X_{t-1}, X_{t-2}, \dots, X_{t-k}.$



SIGNAL ESTIMATION:

$$X_n = Z_n + N_n$$

OBSERVATION STATE NOISE

$$Z_n = A_n Z_{n-1} + w_{n-1}$$

CURRENT STATE PREVIOUS STATE NOISE

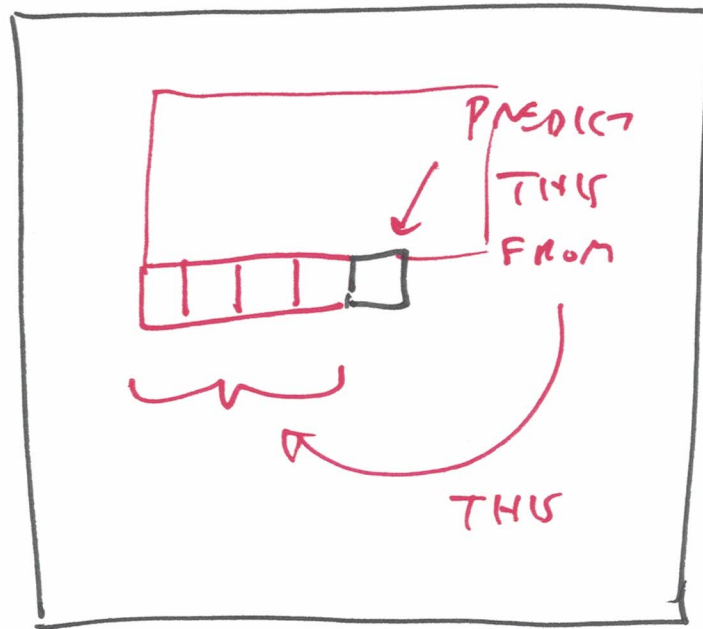
TRANSITION

PREDICT

$$\hat{Z}_n = \sum_{i=1}^k h_i x_i$$

LINEAR COMBINATION OF OBSERVATIONS

PREDICTIVE CODING:



ANY PROBLEM INVOLVING NOISE IS
A PROBABILITY PROBLEM.

4000-LEVEL ~~STRA~~ ML, COMMS

6000-LEVEL ALMOST EVERYTHING

NO CLASS ON MONDAY APR 27!

REVIEW SESSION ON

MONDAY MAY 4 3-4:30 PM
IN CLASSROOM.

FINAL EXAM ON

THURSDAY MAY 7 3-6 PM
IN CLASSROOM

+ THE USUAL OFFICE HOURS